

PROJECT CHARTER

Autonomous Culture & Wellness Intelligence Tool

System Names: PulseIQ
HR and Management Group 02

1. Project Background

Factors such as employee engagement and workplace culture are very important because they determine the performance of an organization, the level of satisfaction among employees, and the dynamics associated with talent retention within an organization. Typically, there have been formal surveys that are used by organizations to gauge the level of employee engagement on a quarterly or biannual basis. Although these traditional methods produce valuable historic data, they often miss out on capturing emerging issues at work because of the delays involved.

The objective of this project is to create a platform for intelligent monitoring of workplace culture and employee engagement through an autonomous, web-based platform. Through the adoption of weekly pulse surveys as opposed to biannual surveys, this platform will be able to track the trajectory of response patterns to identify emerging problems within the organization and give feedbacks in the form of AI-assisted diagnosis.

2. Problem Statement

Structural constraints for modern Human Capital Management activities arise because of the design of conventional frameworks used to facilitate employee feedback mechanisms:

- **Infrequent Collection:** The process becomes lengthy enough to make feedback reactive instead of being proactive.
- **Lagging Visibility:** The problems within the organization's culture or workplace environment are usually detected only when they become retention or productivity risks.
- **Analytical Bottlenecks:** Legacy and manual reporting systems demand substantial turnaround time to sanitize, correlate, and translate collected data into presentable findings.
- **Information Gaps:** Managers in charge of operations do not have access to detailed and timely information on how employees feel about their work.
- **Missed Opportunities:** Companies miss the chance to perform interventions or give support at the moment when the friction arises.

3. Proposed Solution

The recommended solution would be an enterprise-scale, AI-enabled web app that is designed to continually observe and analyze the organizational culture and employee wellness indicators using automated weekly pulse surveys. The main workflow processes would include the following elements:

- **Automated Ingestion:** Gathering minimal effort required weekly feedback loops from distributed teams.
- **Trend Progression Diagnostics:** Monitoring micro and macro trends of operational wellness metrics over defined timeframes.
- **Anomaly & Recurrence Detection:** Leveraging statistical patterns in order to draw attention to recurring negative trends that require immediate attention.
- **Configurable Alert Systems:** Providing deterministic alerts to responsible line managers or HR specialists on a case-by-case basis depending on defined threat parameters.
- **Generative Insight Contextualization:** Leveraging safe Large Language Models (LLMs) to automatically provide context to unstructured textual feedback and generate anonymized executive summaries along with actionable recommendations.

4. Project Objectives

Specific Primary Objective: Conducting research, development, and implementation of an autonomous culture and wellness intelligence platform which converts frequent employee feedback to understandable decision vectors for management.

Operational Objectives:

- Construction of the execution procedure of the weekly pulse survey that is highly intuitive, efficient, and has low friction.
- Building multidimensional visualization models to measure the dynamics of shift metrics across subjective and objective wellness.
- Automation of algorithms to detect recurring systemic problems that could jeopardize business outputs.
- Designing of special role-separated analytics dashboard with real-time computation refreshes.
- Integration of secure AI text summarization technology to process sentiment fields without breaching base identity anonymization safeguards.
- Faster response to human resources lifecycle events through the creation of trigger alerts.

5. Project Scope

In-Scope Features	Out-of-Scope Elements
• Employee Login Verification and Directory Access • Automated Weekly "Pulse" Survey Form Interface • Quantitative Measures of Wellbeing Benchmarking • Optional Free Text Comment Contexts • Anonymous Open-Ended Text Feedback Channels • Role-Based Management Reporting Dashboards • Macro-analytics for HR Enterprise Dashboard • Automated Trend Anomalies Detection Systems • Notifications on Conditions-Based Triggers • AI-Based Qualitative Sentiment Summaries • Exportable Management Reports & Analyses	• Payroll & Compensation Engines • Time & Attendance or Shift Tracking Software • Formal Performance Evaluations or OKR Reviews • Talent Acquisition & Recruiting Tools • Clinical or Telemedicine Diagnoses • Leave/PTO Management Systems

6. Project Stakeholders

- **Employees:** End-users offering week-by-week metric data and cultural insight into sentiment data.
- **Managers/Team Leaders:** Key consumers of local sentiment data and recommendations offered by AI.
- **Human Resources (HR) Department:** Corporate administrators wanting to analyze global organizational metrics.
- **system Administrators:** Technicians in charge of system infrastructure, user access provisioning, and API monitoring.
- **Project Team:** Pair of developers responsible for full-stack software development.
- **Academic Supervisor:** Academic expert evaluating methods used in engineering work.

7. Project Deliverables

- Web Application that is fully functioning and ready to deploy for Enterprise use (both Frontend and Backend code).
- Secure relational database schemas deployed on a production-ready MySQL environment.
- Asynchronous AI Insights engine based on secure endpoints of LLMs.
- Project documentation including system architecture, API documentation, and final report presentations.
- Version-controlled source code repository, production-tested under rigorous workflow standards.
- User manual and Verification presentation slides.

8. Technology Stack Specification

Component	Selected Technology Suite
Frontend Client	React (component-based SPA framework for a responsive UI)
Backend Engine	Node.js + Express (handles REST API requests and business logic)
Database Infrastructure	MySQL (stores team, user, and historical survey data)
Styling & Presentation	Tailwind CSS (utility-first framework for responsive design)
Data Visualization	Chart.js (renders trend charts and dashboards)
Security & Authentication	JWT (JSON Web Tokens) for stateless user sessions
Version Management	GitHub (source control and code review)
Artificial Intelligence Suite	OpenAI API (used for generating text summaries from survey feedback)

9. Risk Management Framework

Identified Risk Parameter	Strategic Mitigation Path
Constrained Development Timeline	Stick to a fixed weekly delivery schedule with clearly scoped Agile milestones.
AI API Resource/Quota Limits	Use fallbacks like local token batching, optimized prompts, or open-source local models if API limits are hit.
Team Coordination Overhead	Keep frontend and backend work clearly separated through an API-first approach, so each side can build independently.
Scope Creep & Volatile Requirements	Freeze the engineering scope right after stakeholders sign off on initial requirements.

10. Project Success Criteria

The platform deployment shall be termed empirically successful in case of meeting the following systemic conditions:

- **Submission Accuracy:** Whether authenticated or unauthenticated, users are able to submit survey data without any problems of losses or misalignment.
- **Analytical Accuracy:** Managers are able to perform calculations and produce real-time graphs from historical team data trends.
- **Capture of Deterministic Events:** The platform algorithm detects predefined thresholds of events and produces notifications for all of them.
- **Notification Orchestration:** Notification mechanisms are able to produce system-wide notifications based on dynamic thresholds.
- **Accuracy of Insights from AI Generative Synthesis Layers:** Produced insights are coherent and accurate representations of the trends in the underlying text data, without hallucinations.